

Alex Rivers

Systems & Infrastructure Engineer (Infrastructure focus)

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Computer Science student specializing in distributed systems, storage engines, and ML training infrastructure. Experienced in building transactional key-value databases, implementing consensus algorithms, and optimizing high-throughput data processing pipelines. Passionate about kernel development, low-latency network programming, and platform automation.

Professional Experience

Systems & Linux Kernel Intern

May 2025 -- Aug 2025

KernelCorp Technologies

San Jose, CA (Remote)

- Designed and implemented an eBPF/XDP-based load-balancer that reduced packet processing latency by 35% compared to IPVS-based routing.
- Patched network driver queues to optimize cache-line utilization, resulting in a 12% increase in packet throughput for 100GbE interfaces.
- Developed a Linux Kernel Module (LKM) for virtual device drivers to mock high-frequency sensors for CI/CD test automation.
- Collaborated with upstream developers to patch a race condition in kernel slab memory allocation under heavy network load.

Distributed Storage Intern

May 2024 -- Aug 2024

StoreScale Systems

Seattle, WA

- Integrated RocksDB as a pluggable storage engine for a custom Raft-based transactional database, improving write throughput by 40%.
- Engineered an asynchronous log-compaction pipeline in Go to reduce tail latency of client writes under disk I/O bottlenecks.
- Built a cluster simulation framework to perform partition testing (Jepsen-style) and validated correctness of linearizable reads.
- Optimized gRPC request-serialization paths using Protobuf custom-marshaling, achieving an 18% reduction in CPU utilization.

Projects

TensorFlow & PyTorch ML Training Orchestrator (Python, PyTorch, Kubernetes, CUDA, gRPC, Docker)

Mar 2025

- Developed a custom Kubernetes controller in Go that schedules distributed PyTorch training jobs across multiple GPU-enabled nodes.
- Configured torch.distributed tensor-parallel setups with custom communication primitives, optimizing NCCL ring-allreduce performance.
- Implemented a dynamic fault-recovery agent that restarts failed worker pods, restoring from latest weights checkpoints in under 30 seconds.
- Designed an active GPU allocation scheduler that schedules training jobs to maximize NVLink interconnect bandwidth between GPUs.

Distributed Raft Consensus Engine (Go, Raft, gRPC, Protobuf, Linearizability, Jepsen)

Dec 2024

- Built a complete Raft consensus module in Go supporting leader election, log replication, and safe cluster state transitions.
- Implemented a lease-based read optimization and read-index queries to support linearizable reads without full roundtrip consensus.
- Designed a custom snapshotting mechanism and log compaction process to reclaim disk space, preventing memory bloat on large clusters.
- Passed Jepsen partition tests, simulating network split-brain scenarios and validating that data remains consistent and serializable.

AetherDB: MVCC Key-Value Engine (C++, LSM-Tree, MVCC, POSIX, WAL, Transactions)

May 2024

- Designed an LSM-Tree storage layout in C++ with MemTable, Write-Ahead Log (WAL), and multi-tiered SSTables for fast writes.
- Implemented a lock-free Multi-Version Concurrency Control (MVCC) system using atomic hardware pointers, ensuring snapshot isolation.
- Programmed custom merge-compaction threads using POSIX APIs, reducing storage footprint by 45% and eliminating compaction spikes.
- Engineered an instant recovery algorithm scanning WAL files, restoring database state in under 150ms after abrupt shutdowns.

Technical Skills

Tools: Git & GitHub Actions, Prometheus & Grafana
Systems & Infrastructure: Linux Kernel, Kubernetes, gRPC & Protobuf, eBPF / XDP, PostgreSQL & Redis, Docker
Languages: Rust, C++, Go, Python, C

Education

Bachelor of Science

Tech Institute of Technology

Sept 2023 -- May 2027

Boston, MA

GPA: 3.95/4.00

- Teaching Assistant: Distributed Systems (CS 452), Operating Systems (CS 321)
- Coursework: Distributed Systems, Database Internals, Compiler Construction, Computer Architecture, Advanced Networks